GAUTHAMKRISHNA E

Mechatronics Engineering Student | Automation & Embedded Systems

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PROFILE

Mechatronics Engineering student (7th semester) with hands-on, team-based experience across embedded systems, IoT, sensor-based automation, and automotive-relevant electronics, gained through academic projects, hackathons, and industrial exposure in instrumentation and process automation. Comfortable working across the hardware-software boundary, from sensor integration and microcontroller programming to real-time monitoring dashboards. Seeking to apply this foundation in automation, embedded systems, and IoT within a growth-oriented engineering role, while continuing to build deeper industry and technical expertise.

EDUCATION

B.E. Mechatronics Engineering — Sona College of Technology, Salem *2023 – 2027*

CGPA: 8.21 / 10

Higher Secondary (+2) — Don Bosco Higher Secondary School | 95% *2022 – 2023*

SSLC (10th) — Hari Sri Vidya Nidhi School | 87% *2020 – 2021*

TECHNICAL SKILLS

- Automation & Controls: PLC Programming fundamentals, Industrial Automation, Sensors & Actuators
- Embedded Systems & IoT: Arduino, ESP32, Embedded C, Sensor Integration
- Electronics & PCB Design: KiCad, Altium, PCB Basics
- 3D Design Software: SolidWorks, Solid Edge
- Programming & Data: Python, C, SQL

PROFESSIONAL SKILLS

Problem Solving • Team Collaboration • Time Management • Adaptability

AREAS OF INTEREST

PLC Automation • PCB Design • Embedded Systems • IoT

PROJECTS

Underground Gas Pipeline Monitoring System

- Contributed to an IoT-based monitoring system tracking flow rate, pressure, and temperature in real time using ESP32 and ThingSpeak
- Gained exposure to end-to-end sensor-to-cloud data pipeline design applicable to industrial process and condition monitoring

Adaptive Visual Comfort Accessory

- Contributed to an ESP32-driven adaptive laptop tilt mechanism combining a servo motor with camera-based vision sensing for automated ergonomic alignment
- Gained exposure to real-time sensing and closed-loop actuation, core principles used in automated adjustment and comfort systems

Fingerprint-Based Car Ignition System

- Contributed to an Arduino-based biometric vehicle ignition system using a fingerprint sensor for secure, keyless access control
- Worked on integrating an L298N motor driver with DC motors to translate authenticated fingerprint input into ignition control, directly relevant to automotive access and security systems

INDUSTRIAL EXPOSURE (IN-PLANT TRAINING)

Fertilisers and Chemicals Travancore Limited (FACT) — Instrumentation Dept.

- Completed an 8-day in-plant training gaining exposure to industrial instrumentation, process monitoring systems, and automation practices in fertilizer manufacturing

Steel Authority of India Limited (SAIL) — Mechanical Department

- Completed a one-week training gaining exposure to steel manufacturing processes and mechanical operations in an industrial environment

HACKATHONS & COMPETITIONS

- SCT Hackfest 2026 — Contributed to building an autonomous battle bot using ultrasonic sensors for obstacle detection
- National Project Competition — Contributed to a drone-based Edge AI system for plant disease detection and automated spraying
- Quantum 25 (Mechatronics Department Symposium) — Participated in the Robo Soccer event

WORKSHOPS

- Drone Boot Camp — Built and programmed a quadcopter; learned drone aerodynamics and control system fundamentals
- NEXUS '24 Serial Manipulator Workshop — Gained exposure to robotic manipulator design, control mechanisms, and basic kinematics

CERTIFICATIONS

Solid Edge Certification (Siemens) • Automation in Manufacturing (NPTEL) • Design Thinking: A Primer (NPTEL)

SQL Certificate (LnD) • IOSH Managing Safely Certification • Python Certificate (IBMCE - CEPYT1N)

LANGUAGES

English • Malayalam • Tamil • Hindi